

2_Numeric_datatypes_and_Strings

September 15, 2026

```
[1]: a=2
      print(a," is of type ", type(a))
```

2 is of type <class 'int'>

```
[2]: a=2.5
      print(a," is of type ", type(a))
```

2.5 is of type <class 'float'>

```
[3]: a=2+3j
      print(a," is of type ", type(a))
      print("a is complex number? ", isinstance(2+3j,complex))
```

(2+3j) is of type <class 'complex'>
a is complex number? True

```
[4]: single_quote='Hanu'
      print(single_quote," is of type ", type(single_quote))
```

Hanu is of type <class 'str'>

```
[5]: double_quote="Hanu"
      print(double_quote," is of type ", type(double_quote))
```

Hanu is of type <class 'str'>

```
[8]: triple_quote="""Hanu
      rao
      allada"""
      print(triple_quote," is of type ", type(triple_quote))
```

Hanu
rao
allada is of type <class 'str'>

```
[9]: mystring='python'
      print(mystring[0]) # prints first character
      print(mystring[1]) # prints second character
      print(mystring[1:4]) # prints characters starting from 2nd to 4th
      print(mystring[2:]) # prints characters starting from 3rd character
```

p
y
yth
thon

```
[10]: string1='Hello'
      string2='World'
      print(string1 + ' ' + string2) # prints Hello World
      concat_string=string1 + ' ' + string2
      print(concat_string) # prints Hello World
```

Hello World
Hello World

```
[12]: string3=10
      concat_string=string1 + ' ' + string2 + ' ' + string3
      print(concat_string)
```

```
-----
TypeError                                Traceback (most recent call last)
/tmp/ipykernel_26650/308332608.py in <cell line: 0>()
      1 string3=10
----> 2 concat_string=string1 + ' ' + string2 + ' ' + string3
      3 print(concat_string)

TypeError: can only concatenate str (not "int") to str
```

```
[13]: string3=10
      concat_string=string1 + ' ' + string2 + ' ' + str(string3)
      print(concat_string)
```

Hello World 10

```
[18]: mystring="      Hello, World      "
      print(len(mystring)) # prints 27
      print(mystring.strip()) # prints Hello World
      print(len(mystring.strip())) # prints Hello World
      print(mystring.lower())
      print(mystring.upper())
      print(mystring.replace("Hello", "Hi"))
      print(mystring.replace("o", "*"))
      print(mystring.strip())
      print(mystring.split(","))
      print(mystring.split(" "))
```

31
Hello, World
12

```
hello, world
HELLO, WORLD
Hi, World
Hell*, W*rld
Hello, World
['      Hello', ' World      ']
['', '', '', '', '', '', '', '', 'Hello,', 'World', '', '', '', '', '', '',
'', '', '', '']
```